Mandarin Tone Coach

Adaptive Pitch Feedback for L2 Mandarin Tone Learning

Milestone 2

Video Link

Bharath Jagadish (bjagadish3@gatech.edu)
CS6460 Educational Technology
Georgia Tech

Milestone 1 Recap — What We Built

The System

Streamlit web app — real-time Mandarin tone practice

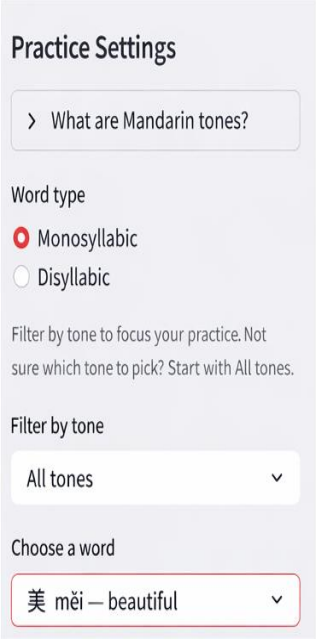
Pipeline: Record → Extract F0 → Classify → Visualize → Feedback

SVM Classifier

- 91.95% speaker-independent accuracy (Tone Perfect corpus, 9,838 MP3s)
- 27 acoustic features: F0 contour, first-order deltas, f0_min_pos
- Main T2↔T3 confusion (16%); rule-based override applied

Practice Modes

- Monosyllabic: 40 words (10 per tone T1–T4)
- Disyllabic: 20 two-syllable words with tone sandhi support
- Pitch contour overlay: learner (red) vs canonical reference (blue)
- Plain-language corrective feedback for all tone-pair combinations



Study Design

Study Protocol

Perception test & Tech. Acceptance Model (TAM) survey

Online, asynchronous, self-paced (~15 min)
No prior Mandarin experience required

1. Pre-test (Form A)

12 audio clips · select the tone heard · no feedback given

2. Practice with the App

Mandarin Tone Coach: monosyllabic & disyllabic words
Real-time pitch contour + plain-language corrective feedback

3. Post-test (Form B)

12 new clips · different speakers

4. TAM Usability Survey

8 Likert items (1–5) + 1 open-ended feedback question

Instruments

Perception Test

- 12 items per form (3 per tone T1–T4)
- Items randomized per participant (seeded by participant ID)
- Scored: proportion correct (0–12 per participant)
- Responses stored in Google Sheets via gspread

Speaker Design

- Form A — Female Voice 1 & Male Voice 1
- Form B — Female Voice 2 & Male Voice 2 (new, unseen speakers)

TAM Survey (Technology Acceptance Model)

- 8 Likert items, 5-point scale (1 = strongly disagree → 5 = strongly agree)
- Constructs: perceived usefulness, ease of use, reuse intent
- 1 open-ended question for qualitative feedback

Participant IDs

- Auto-generated 8-char hex ID shown at end of Form A
- Participants enter their ID to link Form A → Form B

Participant Demographics

13
Paired participants

16
TAM respondents

Prior Mandarin Experience

Never studied before: 11 / 16 (69%)

More than 2 years: 3 / 16 (19%)

Less than 6 months: 2 / 16 (13%)

69% of respondents had no prior Mandarin exposure — confirming the target population of complete novices.

Native Language

English (all variants) 7 / 16 (44%)

Mandarin / Chinese 3 / 16 (19%)

Kannada / Tulu 2 / 16 (13%)

Hindi 1 / 16 (6%)

Farsi 1 / 16 (6%)

Malayalam 1 / 16 (6%)

Swahili 1 / 16 (6%)

Native/heritage Mandarin speakers (n=3) flagged — different performance profile from novices.

Pre/Post Perception Test Results

+10.3 pp

Mean accuracy gain

53.8% → 64.1%

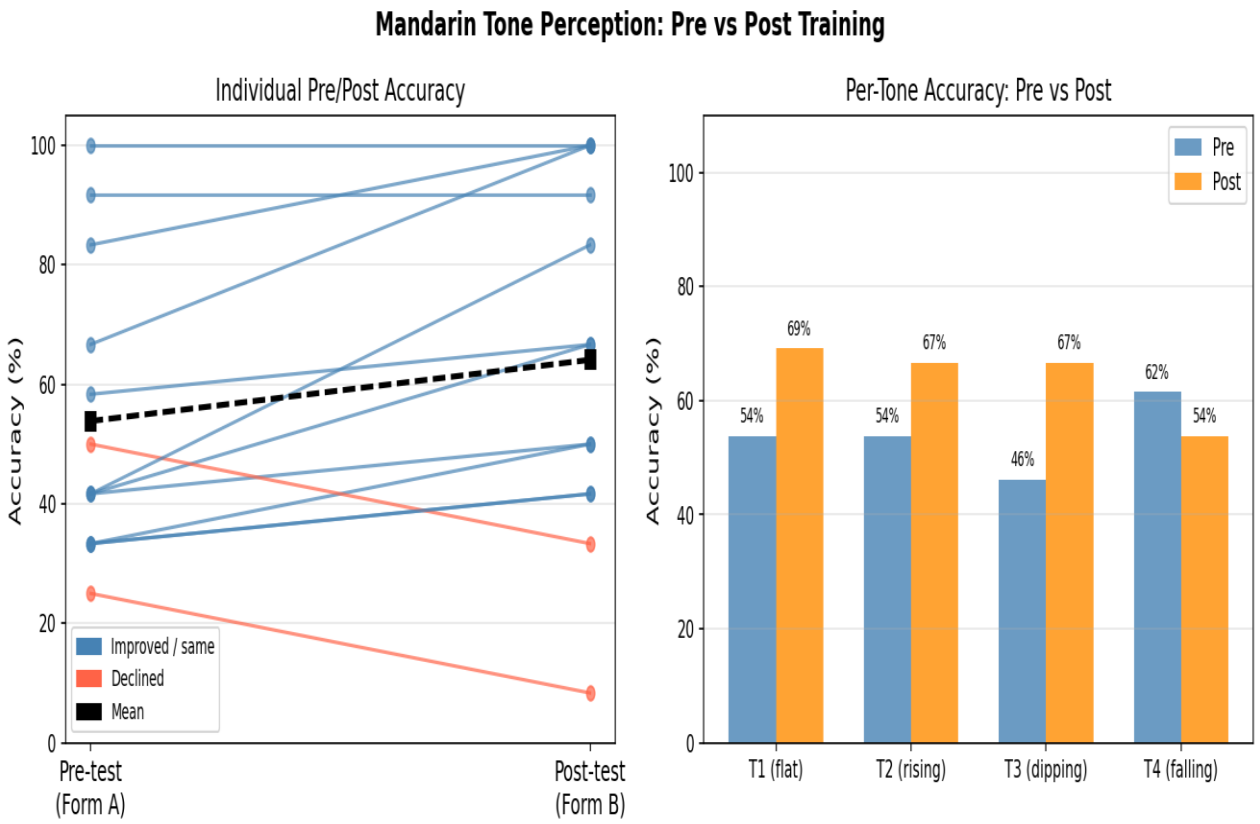
Statistical Tests

Wilcoxon signed-rank test
W = 13.5, p = 0.087 (two-sided)

Paired t-test (reference)
t = −2.17, p = 0.051

n = 13 paired participants
3 items per tone per participant

Interpretation
Strong positive trend; p just above 0.05 threshold — marginal with n=13 (est. power ~35%)

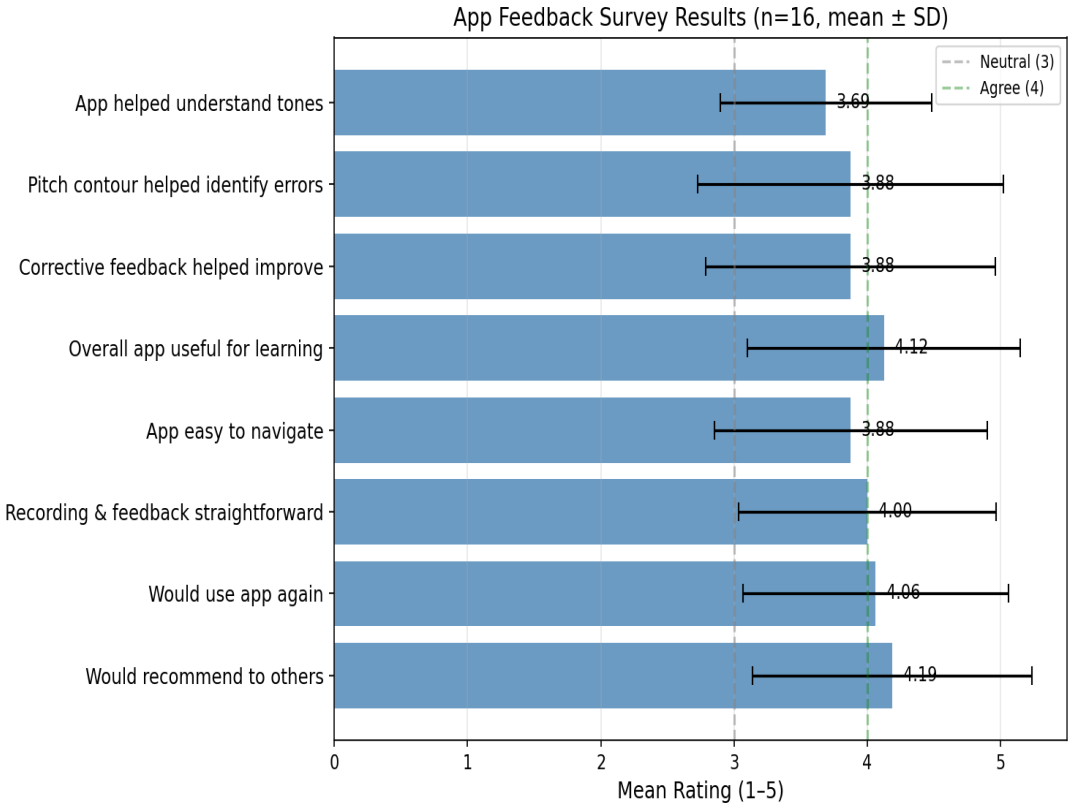


Per-Tone Accuracy

T1 flat	T2 rising	T3 dipping	T4 falling
Pre-test 53.8%	Pre-test 53.8%	Pre-test 46.2%	Pre-test 61.5%
Post-test 69.2%	Post-test 66.7%	Post-test 66.7%	Post-test 53.8%
+15.4 pp ↑	+12.8 pp ↑	+20.5 pp ↑	−7.7 pp ↓

Key observations: T3 (dipping) showed the largest gain (+20.5 pp) — likely benefiting from the pitch contour visualization. T4 declined (−7.7 pp) — possibly harder Form B stimuli or high variance with only 3 items per tone. Caution: n=13, 3 items/tone = limited statistical power.

App Feedback



3.96 / 5.0

Overall mean (8 items)

4.19	Would recommend to others (highest)
4.12	Overall app useful for learning
4.06	Would use app again
4.00	Recording & feedback straightforward
3.69	Helped understand tones (lowest)

⚠ Respondent 13 (native Mandarin speaker who couldn't navigate the app) gave all 1s — excluding this outlier raises the overall mean to 4.09.

Key Findings & Limitations

What Worked

- Positive pre→post trend (+10.3 pp) across all non-native participants
- T3 (dipping) largest gain (+20.5 pp) — pitch contour most helpful
- TAM overall usefulness & reuse intent both 4.12 / 5
- Pitch contour overlay praised in open-ended feedback
- Disyllabic mode broadened practice scope successfully

Participant Feedback

- T2 vs T3 perceptually hard to distinguish — confirmed by test data
- No target audio before recording — users wanted to hear word first
- "Too noisy / too quiet" error not specific enough to act on
- More words requested; 40-word list feels limited
- App navigation confusing for one participant on first load

Limitations

- n = 13 paired — underpowered pilot (est. power ~35%)
- Asynchronous: no control over practice time or noise environment
- T4 decline (−7.7 pp) likely Form B difficulty, not regression
- 3 native speakers in sample — different performance profile
- SVM misclassifies some correct native-speaker productions

"The app is well planned and designed in a way that makes it very easy for the user to learn Mandarin." — Participant 9

What’s Next — Weeks 14–16

Week 14 · Deep Analysis

- ✓ Per-participant learning curve analysis
- ✓ Subgroup: novices vs prior Mandarin learners
- ✓ Perception test confusion matrix (which tones mixed up)
- ✓ Correlate TAM scores with learning gains

Weeks 14–15 · Paper Writing

- ✓ Introduction & literature review (CAPT, Noticing Hypothesis, TAM)
- ✓ Methods: system design + study protocol
- ✓ Results: quantitative pre/post + qualitative survey themes
- ✓ Discussion: implications, limitations, future work

Week 16 · Final Submission

- ✓ Complete paper (course format)
- ✓ Final video walkthrough of the system
- ✓ Open-source release on GitHub
- ✓ Optional: add target-word audio examples to the app